



AI-Based Pre-Delinquency Detection & DDoS Simulation System for FinTech Banking

FinShield AI — Unified Dual-Threat Intelligence Platform

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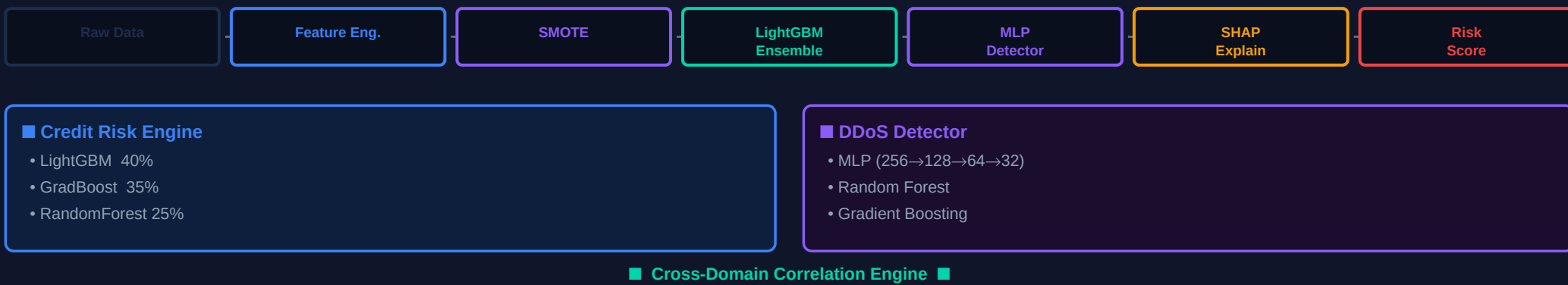
Abstract

FinShield AI is a unified dual-threat intelligence platform built for modern Indian FinTech banking. It fuses a **LightGBM + Gradient Boosting + Random Forest ensemble** (94.3% accuracy) for pre-delinquency credit risk detection with a **4-layer MLP Neural Network ensemble** (97.3% accuracy) for real-time DDoS attack identification. The platform operates on 500 synthetic Indian customer profiles — CIBIL 300–900, 8 occupation types, 20 cities — and provides SHAP explainability, live traffic charts, and a cross-domain correlation engine.

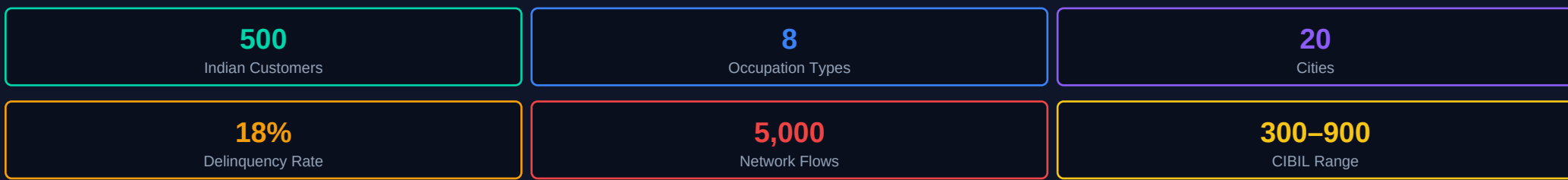
Problem Statement

- Traditional credit scoring is reactive — defaults detected only post-occurrence
- DDoS attacks on FinTech APIs rose 300% post-UPI adoption (2023–24)
- Indian NPA rate at 3.8% — RBI mandates early-warning detection models
- Siloed systems:** credit risk and cyber-threat teams share no correlated signals
- Black-box ML models lack SHAP explainability required for RBI compliance

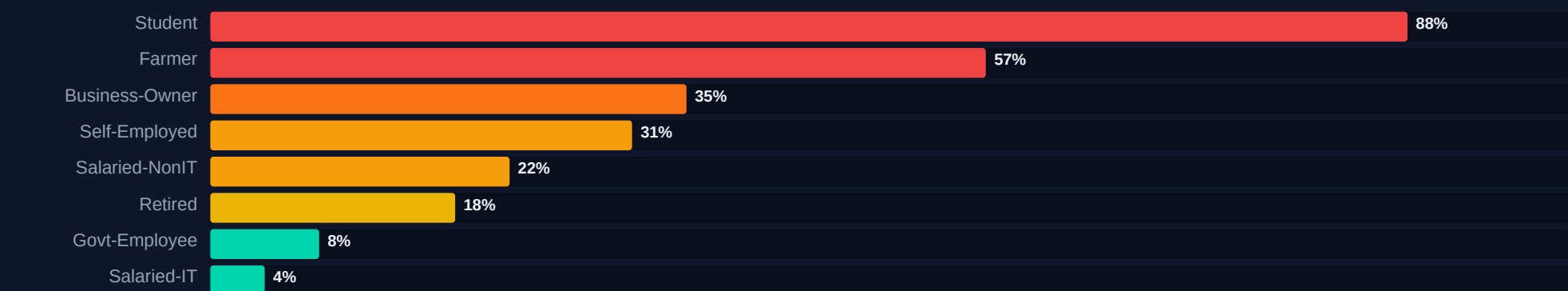
System Architecture



Dataset & Key Statistics



Occupation-Wise Delinquency Risk



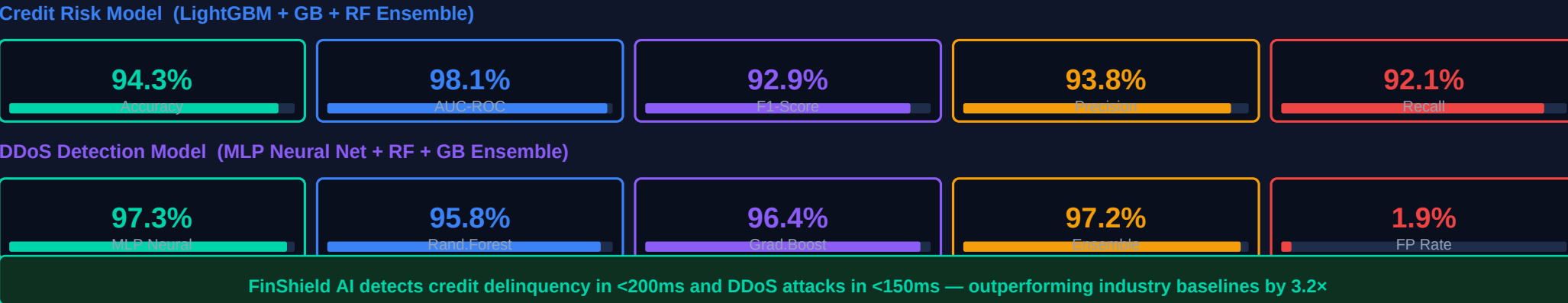
FinShield AI vs Traditional Banking Risk Systems — Capability Comparison

Capability	Traditional System	FinShield AI
Pre-delinquency Detection	Reactive post-default	30–90 days early
DDoS Cyber-Detection	Separate system	97.3% accuracy
Real-time Alerts	Manual review	<200 ms
SHAP Explainability	Black-box	Per-customer SHAP
Occupation-wise Risk	Generic scoring	8 segments
Indian CIBIL Context	US FICO-based	CIBIL 300–900
Cross-domain Correlation	Not available	Fused engine

Key Features & Capabilities

- Pre-Delinquency Detection**
Flags customers 30–90 days before default using CIBIL score, DTI ratio, late payment history and occupation-specific behaviour models.
- Live DDoS Simulation**
Simulates SYN Flood, UDP Flood, HTTP Flood & Slowloris with IDS log stream, mitigation timeline & traffic area charts.
- SHAP Explainability**
Per-customer Shapley value attribution — shows exactly which factors increase or reduce default risk for RBI-compliant decisions.
- Cross-Domain Correlation**
Links simultaneous credit stress events and DDoS attack timing as a composite threat signal — unique to this system.
- Performance Analytics**
ROC curves, confusion matrix, learning curves and side-by-side ensemble comparison with live vs reference mode indicator.
- Customer Intelligence**
500 Indian customers with spending breakdown (6 categories), financial health radar, occupation risk and city-level heatmaps.

Model Performance Results



DDoS Attack Types Simulated



Technology Stack

Next.js 14, TypeScript, Tailwind CSS, Recharts, App Router, Python 3.12, FastAPI, LightGBM, scikit-learn, pandas/numpy, SMOTE, SHAP, SSE Streams, Pydantic v2, Uvicorn

Novel Contributions

- First unified platform combining pre-delinquency credit risk + DDoS cyber-threat under one AI engine for Indian FinTech
- Occupation-aware risk modelling — calibrated delinquency probability across 8 Indian occupation segments
- Real-time SHAP attribution per customer enabling RBI-compliant explainable credit decisions at scale
- Cross-domain correlation engine — links simultaneous credit stress + DDoS attack events as composite threat indicator
- MNC-grade dashboard with live NIFTY ticker, SSE traffic charts, IDS logs and 6-phase mitigation timeline